

FREEDOM INTERNATIONAL SCHOOL
SUBJECT - MATHEMATICS
TOPIC – STRAIGHT LINES

CLASS: XI

NAME

1. A line passes through the point (2,2) and is perpendicular to the line $3x + y = 3$. Its y-intercept will be _____.
a) $\frac{1}{3}$ b) $\frac{2}{3}$ c) 1 d) $\frac{4}{3}$
2. If p be the length of the perpendicular from the origin on the line $\frac{x}{a} + \frac{y}{b} = 1$, then,
a) $p^2 = a^2 + b^2$ b) $p^2 = \frac{1}{a^2} + \frac{1}{b^2}$
c) $\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$ d) none of these
3. The equation of the line passing through (1,5) and perpendicular to the line $3x - 5y + 7 = 0$ is _____.
a) $5x + 3y - 20 = 0$ b) $3x - 5y + 7 = 0$
c) $3x - 5y + 6 = 0$ d) $5x + 3y + 7 = 0$
4. The inclination of the straight line passing through the point (-3,6) and the midpoint of the line joining the point (4, -5) and (-2,9) is _____.
a) $\frac{\pi}{4}$ b) $\frac{\pi}{6}$ c) $\frac{\pi}{3}$ d) $\frac{3\pi}{4}$
5. Distance between the lines $5x + 3y - 7 = 0$ and $15x + 9y + 14 = 0$ is _____.
a) $\frac{35}{\sqrt{34}}$ b) $\frac{1}{3\sqrt{34}}$ c) $\frac{35}{3\sqrt{34}}$ d) $\frac{35}{2\sqrt{34}}$
6. The angle between the lines $2x - y + 3 = 0$ and $x + 2y + 3 = 0$ is _____.
a) 90° b) 60° c) 45° d) 30°
7. The line segment joining the points (-3, -4) and (1, -2) is divided by y-axis in the ratio _____.
a) 1:3 b) 2:3 c) 3:1 d) 3:2
8. Area of the triangle formed by the points $((a + 3)(a + 4), (a + 3))$, $((a + 2)(a + 3), (a + 2))$ and $((a + 1)(a + 2), (a + 1))$ is _____.
a) $25a^2$ b) $5a^2$ c) $24a^2$ d) none of these

9. If A(6,4) and B(2,12) are two points, the slope of a line perpendicular to the line AB is _____.
a) -2 b) 2 c) $\frac{1}{2}$ d) $\frac{-1}{2}$
10. What can be said regarding a line if its slope is negative?
a) θ is an acute angle.
b) θ is an obtuse angle.
c) Either the line is x-axis, or it is parallel to the x-axis.
d) None of the above
11. In which ratio, (3,4) divides the line segment (1,2) and (4,5) internally?
a) 1:2 b) 2:1 c) 3:4 d) 4:3
12. If the angle between two lines is 45° and the slope of one line is $\frac{1}{4}$, which of the following is the possible value of the slope of the other line?
a) $\frac{5}{3}$ b) $\frac{3}{5}$ c) $\frac{-5}{3}$ d) $\frac{4}{5}$
13. If the line joining (1,2) and (7,6) is perpendicular to the line joining (3,4) and (11, x), find x.
a) 12 b) 16 c) -8 d) -16
14. If the slope of a line is 4 and x-intercept made by the line is 2, the equation of the line will be _____.
a) $y = 4x - 8$ b) $y = 4x + 8$
c) $y = 2x + 4$ d) $y = 2x - 4$
15. If P(1,2), Q(3,5) and R(7,9) form a triangle, find the equation of the median through P.
a) $5x - 4y + 3 = 0$ b) $5x + 4y + 3 = 0$
c) $5x - 4y - 3 = 0$ d) $5x + 4y - 3 = 0$